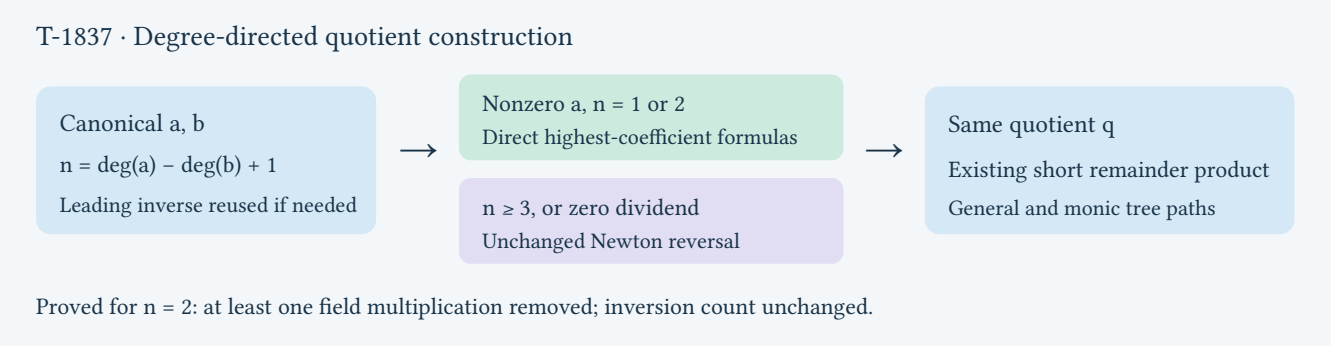
Direct constant and linear quotients

T-1837 · 2026-11-07 · Balanced Euclidean steps

The quotient kernel now reads the leading coefficients directly when the quotient has length one or two. General divrem and monic remainder-tree evaluation share the helper. Larger quotients and zero dividends keep Newton reversal. The same quotient then feeds the existing low remainder product and half-GCD row updates.



Coefficient proof. For equal degrees, q is the leading coefficient ratio. For a one-degree gap, q1=a_m/b_d and q0=(a*(m−1)−q1 b*(d−1))/b_d, taking a missing divisor coefficient as zero. These cancel the highest coefficients exactly. One nonmonic leading inverse is reused; monic scaling and zero cross terms are skipped.

Exact quotient-only cost. Let u flag a nonmonic divisor, h a nonzero subleading divisor coefficient, and ell a nonzero subleading dividend coefficient. The retained kernel and candidate have these field-product counts:

Quotient length	Prior Newton M	Direct M	Either I
1	1	u	u
2	3 + 3h + ell	2u + h	u

Thus a monic constant quotient saves one product, a nonmonic constant quotient ties, and every nonzero linear quotient saves at least one product. Dispatch is a length/nonzero check, not a tuned benchmark threshold. Branches, reductions and allocation costs remain additional; no timing claim follows.

Storage. The direct path avoids reversed input copies, the inverse series and Newton/reversal temporaries. The output quotient, input normalization and low remainder storage remain. This is not a total-memory bound.

Exact software check	Count	Coverage
Constant / linear quotients	224 / 4,744	Exhaustive small leading tuples; cost formulas
Larger quotient controls	640	Same outputs and arithmetic counts
Tree / zero-dividend controls	256 / 8	Evaluation and zero-path preservation
General divisions / resultants	2,592 / 432	Reconstruction; complete half-GCD outputs
Additional recoveries	617	5,929 calls; zero unresolved
Inherited public comparisons	63	511 calls

Evidence and scope. Source 1,108-point/24-window checks and prior singleton, matrix, division-value, lifetime and preparation controls pass. Normal build pins seven native dependencies without audits. Original 4,563-case cohort unchanged; no cloud, timing benchmark, external comparison or full Cargo rerun. PCC-20261107-18740d is unsigned producer evidence. See DERIVATION.md and receipts for commands, assumptions and hashes.

Sources and next work. Primary FLINT division documentation and exact predecessor kernels; KB unavailable. Standard specialization, no novelty claim. Next derive a wider triangular-quotient cost envelope that charges coefficient inspection and temporary storage before extending dispatch.